

MINI PROJECT:

Tasks

1.

Create a console-based app called Music Listening Logger that allows users to enter the number of minutes they listened to music each day for a week and stores this data in an array.

```
#include <stdio.h>
```

```
int main() {
```

```
    int listeningMinutes[7];
```

```
    for (int i = 0; i < 7; i++) {
```

```
        printf("Enter listening minutes for day %d: ", i + 1);
```

```
        scanf("%d", &listeningMinutes[i]);
```

```
    }
```

```
    printf("\nWeekly Listening Data:\n");
```

```
    for (int i = 0; i < 7; i++) {
```

```
        printf("Day %d: %d minutes\n", i + 1, listeningMinutes[i]);
```

```
    }
```

```
    return 0;
```

```
}
```

```
PS C:\Users\SIDDHI\OneDrive\Desktop\c> gcc music_logger_task1.c -o music_logger_task1
PS C:\Users\SIDDHI\OneDrive\Desktop\c> .\music_logger_task1.exe
Enter listening minutes for day 1: 100
Enter listening minutes for day 2: 28
Enter listening minutes for day 3: 37
Enter listening minutes for day 4: 83
Enter listening minutes for day 5: 91
Enter listening minutes for day 6: 78
Enter listening minutes for day 7: 45

Weekly Listening Data:
Day 1: 100 minutes
Day 2: 28 minutes
Day 3: 37 minutes
Day 4: 83 minutes
Day 5: 91 minutes
Day 6: 78 minutes
Day 7: 45 minutes
```

2.

Add a menu-driven interface to your Music Listening Logger so users can choose to log new listening minutes, view their weekly summary, or exit the app.

Hint: Use a loop to repeatedly show the menu until the user selects exit.

```
#include <stdio.h>
```

```
int main() {
    int listeningMinutes[7] = {0};
    int choice;

    while (1) {
        printf("\n==== Music Listening Logger =====\n");
        printf("1. Log Listening Minutes\n");
        printf("2. View Weekly Summary\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
```

```
scanf("%d", &choice);

if (choice == 1) {
    for (int i = 0; i < 7; i++) {
        printf("Enter listening minutes for day %d: ", i + 1);
        scanf("%d", &listeningMinutes[i]);
    }

    printf("Listening data saved successfully.\n");
}

else if (choice == 2) {
    printf("\nWeekly Listening Summary:\n");

    for (int i = 0; i < 7; i++) {
        printf("Day %d: %d minutes\n", i + 1, listeningMinutes[i]);
    }
}

else if (choice == 3) {
    printf("Exiting Music Listening Logger...\n");
    break;
}

else {
    printf("Invalid choice. Please try again.\n");
}
}
```

```
return 0;
```

```
}
```

```
PS C:\Users\SIDDHI\OneDrive\Desktop\c> gcc music_logger_task2.c -o music_logger_task2
```

```
PS C:\Users\SIDDHI\OneDrive\Desktop\c> .\music_logger_task2.exe
```

```
===== Music Listening Logger =====
```

```
1. Log Listening Minutes
```

```
2. View Weekly Summary
```

```
3. Exit
```

```
Enter your choice: 1
```

```
Enter listening minutes for day 1: 10
```

```
Enter listening minutes for day 2: 20
```

```
Enter listening minutes for day 3: 30
```

```
Enter listening minutes for day 4: 20
```

```
Enter listening minutes for day 5: 34
```

```
Enter listening minutes for day 6: 43
```

```
Enter listening minutes for day 7: 84
```

```
Listening data saved successfully.
```

3.

Implement file handling in your Music Listening Logger so that when the user logs their daily listening minutes, the data is saved to a file named music_log.txt for persistence.

```
#include <stdio.h>
```

```
int main() {
```

```
    int listeningMinutes[7];
```

```
    FILE *file;
```

```
    for (int i = 0; i < 7; i++) {
```

```
        printf("Enter listening minutes for day %d: ", i + 1);
```

```
        scanf("%d", &listeningMinutes[i]);
```

```
    }
```

```
    file = fopen("music_log.txt", "w");
```

```

if (file == NULL) {
    printf("Error opening file.\n");
    return 1;
}

for (int i = 0; i < 7; i++) {
    fprintf(file, "Day %d: %d minutes\n", i + 1, listeningMinutes[i]);
}

fclose(file);

printf("\nMusic listening data saved to music_log.txt successfully.\n");

return 0;
}

```

4.

Add a feature to your app that reads the saved music_log.txt file and generates a weekly report showing the total, average, and highest listening minutes in the week.

```
#include <stdio.h>
```

```

int main() {
    FILE *file;
    int minutes;
    int total = 0;
    int count = 0;

```

```
int highest = 0;

file = fopen("music_log.txt", "r");

if (file == NULL) {
    printf("Error opening music_log.txt.\n");
    return 1;
}

while (fscanf(file, "Day %*d: %d minutes", &minutes) == 1) {
    total += minutes;

    if (minutes > highest) {
        highest = minutes;
    }

    count++;
}

fclose(file);

if (count > 0) {
    printf("Weekly Music Listening Report\n");
    printf("Total: %d minutes\n", total);
    printf("Average: %.2f minutes\n", (float)total / count);
    printf("Highest: %d minutes\n", highest);
} else {
    printf("No listening data found.\n");
}
```

```

return 0;
}
PS C:\Users\SIDDHI\OneDrive\Desktop\c> gcc music_logger_task4.c -o music_logger_task4
PS C:\Users\SIDDHI\OneDrive\Desktop\c> .\music_logger_task4.exe
Weekly Music Listening Report
Total: 120 minutes
Average: 120.00 minutes
Highest: 120 minutes

```

5.

Refactor your Music Listening Logger to allow users to reset their weekly data by clearing the array and deleting the contents of music_log.txt.

Constraint: Make sure the reset option is available in the menu and asks for confirmation before deleting data.

```
#include <stdio.h>
```

```

int main() {
    int listeningMinutes[7] = {120, 90, 150, 80, 200, 170, 100};
    int choice;
    char confirmation;

    while (1) {
        printf("\n==== Music Listening Logger =====\n");
        printf("1. View Weekly Data\n");
        printf("2. Reset Weekly Data\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        if (choice == 1) {
            printf("\nWeekly Listening Data:\n");

            for (int i = 0; i < 7; i++) {
                printf("Day %d: %d minutes\n", i + 1, listeningMinutes[i]);
            }
        }
    }
}

```

```

    }
}
else if (choice == 2) {
    printf("Are you sure you want to reset all data? (y/n): ");
    scanf(" %c", &confirmation);

    if (confirmation == 'y' || confirmation == 'Y') {
        for (int i = 0; i < 7; i++) {
            listeningMinutes[i] = 0;
        }

        FILE *file = fopen("music_log.txt", "w");

        if (file != NULL) {
            fclose(file);
        }

        printf("Weekly data has been reset successfully.\n");
    }
    else {
        printf("Reset cancelled.\n");
    }
}
else if (choice == 3) {
    printf("Exiting Music Listening Logger...\n");
    break;
}
else {
    printf("Invalid choice. Please try again.\n");
}
}

return 0;

```

```
}
```

```
PS C:\Users\SIDDHI\OneDrive\Desktop\c> gcc music_logger_task5.c -o music_logger_task5
PS C:\Users\SIDDHI\OneDrive\Desktop\c> .\music_logger_task5.exe
```

```
===== Music Listening Logger =====
```

1. View Weekly Data
2. Reset Weekly Data
3. Exit

Enter your choice: 1

Weekly Listening Data:

Day 1: 120 minutes
Day 2: 90 minutes
Day 3: 150 minutes
Day 4: 80 minutes
Day 5: 200 minutes
Day 6: 170 minutes
Day 7: 100 minutes

```
===== Music Listening Logger =====
```

1. View Weekly Data
2. Reset Weekly Data
3. Exit

Enter your choice: 3

Exiting Music Listening Logger...